Stage 1
AI as Conversational Assistant

Stage 2
AI as Reactive Operator

Stage 3
AI as Adaptive Coordinator

Stage 4
AI as Self-Governing Production System

Nascent Diffusion

Growing Industries

Embedded execution · dual-mechanism gains · accountability-bounded

Trade

Progression

1. Document Processing
2. Compliance Assistance
3. Agentic Workflow
4. Business Operation Simulation

Efficiency & Expansion

- ✓ **Automated Data Entry**
- ✓ **Procurement and inventory plan**

Challenge

- **Compliance and Risk Control:** Decisions must follow strict regulatory rules
- **Reliability in Execution:** Errors can propagate across interconnected systems

Media

Progression

1. Structured Generation
2. Tool-using verification
3. Multi-Agent Content Production
4. Towards Multimedia..

Efficiency & Expansion

- ✓ **Faster news drafting and checking**
- ✓ **Script to final content**

Challenge

- **Reliability Bottleneck:** Errors compound across steps
- **Editorial Accountability:** Human oversight still required

Education

Progression

1. Generic tutoring
2. Pedagogical Agents
3. Multi-Agent Class Simulation
4. Evolving Learning Systems

Efficiency & Expansion

- ✓ **Consistent Guidance**
- ✓ **Simulated Classrooms and method**

Challenge

- **Pedagogical Misalignment:** Incorrect teaching strategies
- **Bias and Fairness:** Unequal or misleading feedback

Law

Progression

1. General Legal Assistance
2. Domain-Specific Legal Case Analysis
3. Multi-step argumentation Agent
4. Multi-Agent Legal Systems

Efficiency & Expansion

- ✓ **Litigation Documents Drafting**
- ✓ **Legal Process Simulation**

Challenge

- **High-Stakes Reliability:** Minor errors can lead to severe consequences
- **Accountability Gap:** Legal responsibility for AI outputs is unclear